KXP 2290 Transponder System – General Data

1. General

The Mode S Transponder (KXP 2290) (Mod 6/2049) (Pre Mod 6/2354) replies to Air Traffic Control Radar Beacon System (ATCRBS) Mode A and C, Intermode, and Mode S interrogations. Provisions have been made for the KXP 2290 Transponder System, this is part of the IFR operational bundle customer order option 6209.

Mod 6/2354 upgrades the dual KXP 2290 Transponders to dual KXP 2290A Transponders with ADS-B Out capability. S.O.O. 6347 enables the ADS-B Out functionality.

Each transponder system includes the KXP 2290 or KXP 2290A LRU located behind the associated PFD Controller and bottom and top (diversity) antennas.

For aircraft with S.O.O. 6388 incorporated, there is one KXP 2290A Transponder installed.

For aircraft with S.O.O. 6388 incorporated, a second KXP 2290A Transponder and associated systems are installed on the aircraft by incorporating S.O.O. 6381.

2. Description

A. KXP 2290 Transponder System

1) Transponder System (KXP 2290) (Mod 6/2049) (Pre Mod 6/2354)

A KXP 2290 Mode S transponder is installed behind each pilot's PFD controller. The KXP 2290 has the ability to transmit an 8 digit alphanumeric Flight ID code. The flight crew may enter Flight ID information into the system via the bezel buttons, PFD controller, or MF controller. If the Flight ID field is left blank, the aircraft registration will automatically be transmitted. The aircraft ICAO address is also always transmitted regardless if Flight ID is manually entered by the pilot or not.

The XPDR detail window is displayed in the radio tuning window by pressing the adjacent bezel button or XPDR button on MF Controller and then pressing the DETAIL button on either the PFD or MF Controller. The detail window of the transponder contains soft keys for the selection of XPDR MODE, TCAS functionality and default VFR CODE. The XPDR SEL soft key is used for active transponder selection. The non-selected transponder will be in a standby mode.

The transponders receive input from the aircraft air ground monitor and use this input to automatically switch from ground to flight mode and vice-versa. Standby Mode is normally only used for maintenance purposes. In the event of failure of the air-ground monitor, automatic switching of the transponder will not be provided. When in GND mode, the transponder will inhibit ATCRBS, ATCRBS/Mode S All Call and Mode S-only All Call replies. However, the unit will continue to generate Mode S transmissions when replying to discretely addressed Mode S interrogations.

The transponders support Mode S enhanced surveillance (EHS) requirements by broadcasting additional downlink aircraft parameters (DAP) using a 1090 MHz extended squitter. This functionality is automatic and transparent to the pilot. No control is given or needed. DAP information transmitted by the transponder includes flight ID, magnetic heading, indicated airspeed, true airspeed, groundspeed, vertical speed, bank angle, selected altitude, and altimeter setting.

The transponders use ARINC 735A TCAS interface protocol to interface with the (optional) TCAS II system. As a peripheral to the TCAS II processor, the transponder receives and replies to short and long air to air surveillance and TCAS coordination interrogations, and receives and replies to ground to air surveillance and Comm A interrogations directed to TCAS. When the transponder system transmits, a suppression pulse is sent to the DME and the TCAS to prevent operation of these systems at the same time, which may cause interference.

TCAS traffic can be viewed by the pilot while the aircraft is on the ground when TA or TA/RA mode has been selected (TA/RA mode may be selected, but only TA will be displayed on the ground and during

fight below 1100 feet AGL). The aircraft does not reply to interrogations from other aircraft when in ground mode, but does display information received from aircraft broadcasting Mode C or Mode S transmissions.

XPDR 1 and XPDR 2 (if installed) bottom antenna(s) are installed on the bottom of the fuselage at stations FS103.8 and the top antenna(s) are installed on the top of the fuselage at stations FS186.5. The transponder antennas are a blade shape and have a low contour with a connector on the base.

28 Volt power is protected by a 5 Amp XPDR 1 (Left Bus) and XPDR 2 (Right Bus) (if installed) circuit breaker located on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

The KXP 2290 transponder system interfaces to the following aircraft systems:

a) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

b) Global Positioning System (GPS) (refer to 34-56-00, Global Positioning System (GPS) – General Data).

c) A suppression pulse to/from the KN 63 DME system (refer to 34-54-00, Distance Measuring Equipment (DME) (Mod 6/2049) – General Data).

d) A suppression pulse to/from the TCAS (refer to 34-43-00, Traffic Alert and Collision Avoidance System I (TCAS I) – General Data and 34-43-01, Traffic Alert and Collision Avoidance System II (TCAS II) (Mod S.O.O. 6219) – General Data).

2) Mode S Transponder System (KXP 2290A) with ADS-B Out (Post Mod 6/2354)

Aircraft incorporating Mod 6/2354, have the KXP 2290A Mode S transponder with Automatic Dependent Surveillance – Broadcast (ADS-B) Out functionality. The KXP 2290A transponder is a direct replacement for the KXP 2290 LRU.

The ADS-B out functionality is enabled through the incorporation of customer standard option S.O.O. 6347. Mode 6/2304 SBAS GPS Receivers is a pre-requisite for S.O.O. 6347.

For aircraft with S.O.O. 6388 incorporated, there is one KXP 2290A Transponder installed.

For aircraft with S.O.O. 6388 incorporated, a second KXP 2290A Transponder and associated systems are installed on the aircraft by incorporating S.O.O. 6381.

When the aircraft is on the ground and **GND** is displayed, as long as the transponder is not in STBY mode, the transponder replies to interrogations that are specifically addressed to the aircraft by name. Otherwise, it does not emit transmissions. ADS-B ground stations and ground-based multilateration systems, which are used to monitor aircraft movement on taxiways, are examples of new ATC technologies that have the ability to address interrogations to specific aircraft while they are on the ground.

ADS-B Out status is displayed in the XPDR radio window. The ADS-B Out status displayed in white indicates normal ADS-B Out operation. The ADS-B Out status dispalyed in amber indicates ADS-B Out is failed. The ADS-B Out status not displayed at all indicates ADS-B Out is disabled by Apex configuration option, or the transponder indicates ADS-B Out is selected off, or the transponder is failed.

3. Adjustment/Test

A. Transponder System (KXP 2290A) ADS-B Out Conformity Verification (Post Mod 6/2354 with S.O.O. 6347 for ADS-B Out Enabled)

1) General

The following procedures are for the ADS-B Out conformity verification of the transponder, GPS and ADAHRS systems.

2) Transponder System (KXP 2290A) ADS-B Out Conformity Verification

a) Ensure the aircraft power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the forward avionic shelves, through the forward baggage compartment.

c) Remove top, middle and bottom covers attached to bulkhead station 44.0 for full access.

d) Verify the part number of GPS 1, GPS 2 (if installed) and ADAHRS as follows:

KGS 200 SBAS GPS – P/N 066–01201–0105
KSG 7200 ADAHRS – P/N 065–00188–5103 or 065–00188–5104

e) Reinstall top, middle and bottom covers attached to bulkhead station 44.0 at the aft wall of the baggage compartment. Ensure covers are mounted correctly and secure.

f) Gain access to XPDR 1 and XPDR 2 (if installed) dataplates by removing pilot and co-pilot Primary Flight Display controllers, respectively. (Refer to 31-13-00, Apex Controllers – General Data, Para 4., Removal and Installation).

g) Verify part number of the XPDR 1 and XPDR 2 (if installed) as follows:

KXP 2290A – P/N 066–01198–0102

h) Reinstall Primary Flight Display controllers that were removed to gain access to the transponders unit. (Refer to 31-13-00, Apex Controllers – General Data, Para 4., Removal and Installation).

i) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

j) Wait for the aircraft electrical systems to initialize and the system flags to clear.

k) Verify Primus Apex® system is utilizing Software Load 4.4 indicated by Top Level System P/N EB60002734–0104, displayed in the System Config ID Window on the Systems MFD.

l) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Functional Test Transponder System

1) General

Note It is recommended to coordinate with local ATC before broadcasting with transponder at simulated altitudes to prevent being a source of interference to ATC or TCAS or ADS-B IN equipped aircraft in the area.

Note Transponder ramp test units must have capability to test all parameters listed in the procedure below. For aircraft incorporating S.O.O. 6347, a transponder ramp test unit with the capability to test ADS-B Out parameters is required.

The following procedures are for the functional test of the Transponder 1 and Transponder 2 (if installed) systems.

2) Function Test (KXP 2290 (Mod 6/2049, Pre Mod 6/2354) or KXP 2290A (Post Mod 6/2354) without S.O.O. 6347 for ADS-B Out Not Enabled)

   a) Verify pitot heat is OFF and connect the Pitot-Static System. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test.)

   b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

   c) Wait several minutes for the aircraft electrical and avionics systems to initialize.

   d) Ensure that XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus), circuit breaker on the avionics circuit breaker panel are closed.

   e) Ensure L/R selection on Flight Controller set to “L” .

   f) Ensure ADAHRS 1 is displayed on LH PFD and ADAHRS 2 is displayed on RH PFD.

   g) Set standard barometric pressure (29.92 inHg or 1013 HPa) on PFDs.

   h) Set Mode A code to the appropriate VFR code. For example, 1200 (Canada/U.S.) or 7000 (Europe).

   i) Ensure transponder set to ALT or TA to enable Modes A, C and S.

   j) Set up transponder ramp test unit to interrogate the bottom transponder antenna, in accordance with the instructions in the test set manual.

   k) Install antenna shield boot on the transponder antenna not under test. For example: If testing XPDR 1 bottom antenna, ensure antenna shield boot is on the XPDR 1 top antenna. If testing XPDR 1 top antenna, ensure antenna shield boot is on the XPDR 1 bottom antenna.

   l) Increase airspeed on the Pitot-Static Test set to 60 Knots.

   m) Verify in the XPDR radio window and on the transponder ramp test unit, that the transponder switches from ground to air mode.

   n) Using a transponder ramp test unit, do a transponder Mode S Auto Test, or equivalent.

   o) Ensure the result of the transponder auto test is PASS. Verify the following:

Table 1 Top and Bottom Antenna Parameters

Parameter	Required Value
Effective Radiated Power (ERP)	51.0 – 57.0 dBm or 125 – 500 W or 21.0 – 27.0 dBw
Minimum Threshold Level (MTL)	–74.0 dBm (±3 dBm)
Radio Reply Frequency	1090 Mhz (±1 Mhz)
All-call Address (AA) (24–bit ICAO Address)	Matches Aircraft Address
Tail Number	Matches Aircraft Tail Number
Flight ID	As entered in FMS Init Tab
A CODE	Matches Mode A code set in Para 3.B.2).h)

Table 1 Top and Bottom Antenna Parameters (continued)

Parameter	Required Value
C ALT	Matches LH PFD, rounded to the nearest 100 ft
S ALT	Matches LH PFD, rounded to the nearest 25 ft
Vertical Status (VS)	IN AIR
Squitter	Verify transponder is transmitting squitters
Diversity Isolation	>25 dB

Table 2 Enhanced Surveillance Parameters

Parameter	Required Value
Roll Angle	Matches LH PFD (±0.5°)
Ground Speed	0 kts (±2 kts)
Indicated Airspeed (IAS)	Matches LH PFD (±2 kts)
True Airspeed (TAS)	Matches LH PFD (±2 kts)
Magnetic Heading	Matches LH PFD (±1.0°)
Baro Alt rate	Matches LH PFD (±50 fpm)

p) Ensure the following Common Usage GICB Capability Reports are shown under the Elementary Surveillance BDS 1,7 Test:

Table 3 BDS 1,7 Common Usage GICB Capability Reports

2,0 (Flight ID)
4,0 (Aircraft Vertical Intention)
5,0 (Track and Turn)
6,0 (Heading and Speed Report)

q) Relocate antenna shield boot from top antenna to bottom antenna.

r) Repeat Para 3.B.2).n) and Para 3.B.2).o), Table 1 — Top and Bottom Antenna Parameters for the top antenna. Table 2 — Enhanced Surveillance Parameters is not required to be repeated for the top antenna.

s) Vent Pitot Static Test set to ambient conditions.

t) Verify in the XPDR radio window and on the transponder ramp test unit that the transponder switches from air to ground mode.

u) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

v) Remove antenna shield boot and all test equipment form aircraft.

w) Disconnect the Pitot-Static System (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test).

3) Function Test (KXP 2290A (Post Mod 6/2354) with S.O.O. 6347 for ADS-B Out Enabled)

Note For procedures involving verification of Transponder ADS-B GPS input, ensure aircraft is located such that the GPS receivers can acquire satellite signals.

a) Verify pitot heat is OFF and connect the Pitot-Static System (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test.)

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical and avionics systems to initialize.

d) Ensure that XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus), circuit breaker on the avionics circuit breaker panel are closed.

e) Ensure L/R selection on Flight Controller set to “L” .

f) Ensure ADAHRS 1 is displayed on LH PFD and ADAHRS 2 is displayed on RH PFD.

g) Set standard barometric pressure (29.92 inHg or 1013 HPa) on PFDs.

h) Set Mode A code to the appropriate VFR code. For example, 1200 (Canada/U.S.) or 7000 (Europe).

i) Ensure transponder set to ALT or TA to enable Modes A, C and S.

j) Ensure ADS-B Status indication in the XPDR radio window is white.

k) Set up transponder ramp test unit to interrogate the bottom transponder antenna, in accordance with the instructions in the test set manual.

l) Install antenna shield boot on the transponder antenna not under test. For example: If testing XPDR 1 bottom antenna, ensure antenna shield boot is on the XPDR 1 top antenna. If testing XPDR 1 top antenna, ensure antenna shield boot is on the XPDR 1 bottom antenna.

m) Increase airspeed on the Pitot-Static Test set to 60 Knots.

n) Verify in the XPDR radio window and on the transponder ramp test unit, that the transponder switches from ground to air mode.

o) Using a transponder ramp test unit, do a transponder Mode S auto test, or equivalent.

p) Ensure the result of the transponder auto test is PASS. Verify the following:

Table 4 Top and Bottom Antenna Parameters

Parameter	Required Value
Effective Radiated Power (ERP)	51.0 – 57.0 dBm or 125 – 500 W or 21.0 – 27.0 dBw
Minimum Threshold Level (MTL)	–74.0 dBm (±3 dBm)
Radio Reply Frequency	1090 Mhz (±1 Mhz)
All-call Address (AA) (24–bit ICAO Address)	Matches ICAO address assigned to aircraft
Tail Number	Matches Aircraft Tail Number
Flight ID	As entered in FMS Init Tab

Table 4 Top and Bottom Antenna Parameters (continued)

Parameter	Required Value
A CODE	Matches Mode A code set in Para 3.B.3).h)
C ALT	Matches LH PFD, rounded to the nearest 100 ft
S ALT	Matches LH PFD, rounded to the nearest 25 ft
Vertical Status (VS)	IN AIR
Squitter	Verify transponder is transmitting squitters
Diversity Isolation	>25 dB

Table 5 Enhanced Surveillance Parameters

Parameter	Required Value
Roll Angle	Matches LH PFD (±0.5°)
Ground Speed	0 kts (±2 kts)
Indicated Airspeed (IAS)	Matches LH PFD (±2 kts)
True Airspeed (TAS)	Matches LH PFD (±2 kts)
Magnetic Heading	Matches LH PFD (±1.0°)
Baro Alt rate	Matches LH PFD (±50 fpm)

q) Ensure the following Common Usage GICB Capability Reports are shown under the Elementary Surveillance BDS 1,7 Test:

Table 6 BDS 1,7 Common Usage GICB Capability Reports

0,5 (Airborne Position)
0,6 (Surface Position)
0,7 (Status)
0,8 (Type and Identification)
0,9 (Airborne Velocity Information)
2,0 (Flight ID)
4,0 (Aircraft Vertical Intention)
5,0 (Track and Turn)
6,0 (Heading and Speed Report)

r) Using a transponder ramp test unit, do ADS-B AC 20-165 auto tests, or equivalent.

s) Ensure the result of the ADS-B auto test is Pass. Verify the following:

Table 7 ADS-B Parameters

Parameter	Required Value
Aircraft Position (Lat/Long)	Matches FMS Lat/Long
Aircraft Position Error	Within NACP limits
NACP	≥ 8 EPU < 92.6m (0.05 nm)
NIC	≥ 7 Rc < 370.4m (0.2 nm)
NACV	≥ 1 (<10 m/s) Note Due to the way the KGS 200 GPS units process and filter the NACV value, it is not uncommon for the NACV parameter to display '0' during ground test. This is not considered a fail of the test procedure provided the other GPS derived parameters meet or exceed the required values shown.
E-W Velocity	0
N-S Velocity	0
ADS-B Baro Altitude	Matches LH PFD, rounded to the nearest 25ft
XPDR Baro Altitude	Matches LH PFD, rounded to the nearest 25ft
Geometric Altitude	Matches airfield height ± 100 feet
SIL	≥ 3 (≤$1x10^{-7}$ per hour)
SDA	≥ 2 (≤$1x10^{-5}$ per hour)
TCAS Operational (This parameter is applicable for TCAS II equipped aircraft only)	0 (TCAS II in TA Only Mode) 1 (TCAS II in TA/RA Mode)
RA Active (This parameter is applicable for TCAS II equipped aircraft only)	Matches TCAS II RA status (0 = No RA alert; 1 = Active RA alert)
Mode 3/A Code	As entered in XPDR radio window
Ident	Matches status of Ident (Yes = active)
ICAO Address	Matches ICAO address assigned to aircraft
Emerg/Prior Code	0
Emitter Category	Light
ADS-B In Capability	0 = Not capable
Flight ID	As entered in FMS Init Tab
Aircraft Length/Width Note Available during AC 20–165 Surface Test only.	2 = < 25 m, < 28.5 m

t) Relocate antenna shield boot from top antenna to bottom antenna.

u) Repeat Para 3.B.3).o) and Para 3.B.3).p), Table 4 — Top and Bottom Antenna Parameters for the top antenna. Table 5 — Enhanced Surveillance Parameters is not required to be repeated for the top antenna.

v) Open GPS 1 (J3) (Left Bus) circuit breaker, wait a few seconds for the GPS 1 Fail CAS message to be displayed. Ensure ADS-B status indication remains white, leave GPS 1 circuit breaker open for the next step (dual GPS). Ensure ADS-B status indication turns amber, skip the next step (single GPS).

w) Open GPS 2 (J4) (Right Bus) circuit breaker, wait a few seconds for the GPS 2 Fail CAS message to be displayed, ensure ADS-B status indication turns amber.

x) Close GPS 1 (J3) (Left Bus) circuit breaker, allow enough time for GPS 1 to initialize, ensure ADS-B status indicator returns to white.

y) Close GPS 2 (J4) (Right Bus) circuit breaker (dual GPS).

z) Turn ADS-B Out OFF in the XPDR detail window, ensure ADS-B status indicator clears.

aa) Vent Pitot Static Test set to ambient conditions.

ab) Verify in the XPDR radio window and on the transponder ramp test unit that the transponder switches from air to ground mode.

ac) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

ad) Disconnect the Pitot-Static System (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test.).

ae) Remove antenna shield boot and all remaining test equipment from aircraft.

C. Transponder Air Data Input Verification (KXP 2290 (Mod 6/2049) (Pre Mod 6/2354) or KXP 2290A (Post Mod 6/2354))

1) General

The following procedures verify the air data inputs to the Transponder 1 and Transponder 2 (if installed) systems.

2) Transponder Air Data Input Verification

Note It is recommended to coordinate with local ATC before broadcasting with transponder at simulated altitudes to prevent being a source of interference to ATC or TCAS or ADS-B IN equipped aircraft in the area.

a) Ensure that the (ESIS) STATIC SOURCE valve is in the NORMAL position.

b) Verify pitot heat is OFF and connect a calibrated Pitot/Static Test Set to the Static ports on the aircraft (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test.).

c) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

d) Ensure that XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus), circuit breaker on the avionics circuit breaker panel are closed.

e) Wait for the aircraft electrical system to initialize and the attitude flags to clear.

f) Ensure L/R selection on Flight Controller set to “L” (Post Mod 6/2303).

g) Ensure ADAHRS 1 is displayed on LH PFD and ADAHRS 2 is displayed on RH PFD.

h) Set standard barometric pressure (29.92 inHg or 1013 HPa) on PFDs.

i) Set Mode A code to the appropriate VFR code. For example, 1200 (Canada/U.S.) or 7000 (Europe).

j) Select XPDR 1 for use (dual transponder).

k) Ensure transponder set to ALT or TA to enable Modes A, C and S.

l) Leak test the Static System (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test).

m) Increase airspeed on the Pitot-Static Test set to 60 Knots.

n) Complete the Air Data Verification. Refer to Table 8 — Air Data Verification Table.

Table 8 Air Data Verification Table

LH PFD Altitude (feet)	Mode C Altitude	Mode S Altitude
–1,000	Matches LH PFD Altitude, rounded to the nearest 100 ft.	Matches LH PFD Altitude, rounded to the nearest 25 ft.
0		
525		
1,050		
1,575		
2,100		
3,000		
4,000		
5,000		
6,000		
8,000		
10,000		
12,000		
14,000		
16,000		
18,000		
20,000		
22,000		
25,000		

o) Select XPDR 2 for use (dual transponder).

p) Select ADAHRS 2 on LH PFD and repeat above procedure for Transponder 2 (dual transponder).

q) Vent Pitot Static Test set to ambient conditions.

r) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Part 2.).

s) Disconnect the Pitot/Static Test Set from the static ports on the aircraft (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test).

t) Ensure any tape has been removed and pitot and static ports are clear.

D. Electrical Bonding Test – Upper Transponder 1 and 2 (if installed) Antennas

1) General

The following procedures are for the electrical bonding for the Upper Transponder 1 and 2 (if installed) Antennas.

2) Electrical Bonding Test

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the ceiling compartment aft of station 180.75. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data.

c) Verify that the electrical conductivity is consistent over the antenna electrical bond contact area by placing a probe on the Upper Transponder antenna mounting surface area between the fuselage thru hole and the antenna connector. Place the second probe on the nearest airframe ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise. If necessary, remove antenna connector for additional access to the antenna bonding surface.

Note If disconnecting antenna connectors, do a functional test on the Transponder System, refer to Para 3.B.

d) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

e) Replace the ceiling compartment interior aft of station 180.75. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

E. Electrical Bonding Test – Lower Transponder 1 and 2 (if installed) Antennas

1) General

The following procedures are for the electrical bonding for the Lower Transponder 1 and 2 (if installed) Antennas.

2) Electrical Bonding Test

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to left-hand Access and Inspection Provisions 15 panel. (Refer to 12-00-00, Servicing – General Data).

c) Release fasteners and swing open access panel.

d) Locate Lower Transponder antenna aft of station 101.0.

e) Verify that the electrical conductivity is consistent over the antenna electrical bond contact area by placing a probe on the Lower Transponder antenna mounting surface area between the fuselage thru hole and the antenna connector. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise. If necessary, remove antenna connector for additional access to the antenna bonding surface.

Note If disconnecting antenna connectors, do a functional test on the Transponder System, refer to Para 3.B.

f) If the electrical conductivity does not meet requirements, check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

g) Close left-hand Access and Inspection Provision 15 panel and tighten the fasteners.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECT OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELDS GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELLS/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of Transponder System (KXP 2290 (Mod 6/2049) (Pre Mod 6/2354) or KXP 2290A (Post Mod 6/2354))

1) General

These procedures are for the removal and installation of the XPDR 1 and/or the XPDR 2 (if installed) system.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Gain access to XPDR 1, by removing pilot's Primary Flight Display controller. To gain access to XPDR 2 (if installed), remove co-pilot's Primary Flight Display controller. (Refer to 31-13-00, Apex Controllers – General Data, Para 4.).

d) Using an Allen wrench, turn the securing rod counterclockwise until the unit disengages from the mounting tray.

e) Remove the transponder unit from the mounting rack.

f) Visually inspect mounting tray for condition, if damaged replace as per Para 4.B.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus) circuit breakers are pulled and fitted with safety clips.

c) Gain access to XPDR 1, by removing pilot's Primary Flight Display controller. To gain access to XPDR 2 (if installed), remove co-pilot's Primary Flight Display controller. (Refer to 31-13-00, Apex Controllers – General Data, Para 4.).

d) Slide the transponder unit into the tray until the securing rod engages the nut at the back of the rack.

e) Using an Allen wrench, turn the securing rod clockwise until it has drawn the unit into the rack and is tight.

f) Reinstall Primary Flight Display controller that was removed to gain access to the transponder unit. (Refer to 31-00-00, Instruments – General Data).

g) Remove safety clips and reset XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus) circuit breakers on the avionics circuit breaker panel.

h) Do a function test of the Transponder System, refer to Para 3.B.

B. Removal and Installation of Transponder System Mounting Tray

1) General

These procedures are for the removal and installation of the XPDR 1 mounting tray and the XPDR 2 mounting tray, if installed. The KXP 2290 Transponder shares a mounting tray with the KMC 2210 controller and is part of the APEX cockpit panel configuration.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove the desired PFD Controller and transponder as per Para 4.A.2).

c) Remove screws securing the mounting tray to the instrument panel.

d) Remove electrical connector and coaxial cable from mounting tray.

e) Remove mounting tray from aircraft.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Install electrical connector and coaxial cable to mounting tray.

c) Align the tray to the mounting holes on the instrument panel.

d) Install the screws that secure the mounting tray to the instrument panel.

e) Install the PFD Controller and transponder, refer to Para 4.A.3).

C. Removal and Installation of Transponder Antennas (CI 110–41)

1) General

These procedures are for the removal and installation of the XPDR 1 and XPDR 2 (if installed) antennas.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna and mounting screws.

d) Remove the four screws from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

g) Remove antenna from fuselage.

Note Stow the connectors of the removed components.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus) circuit breakers are pulled and fitted with safety clips.

c) Solvent clean the electrical contact bond areas immediately before the installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with four mounting screws.

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) Touch up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data).

j) Remove safety clips and reset XPDR 1 (G3) (Left Bus) and/or XPDR 2 (G4) (Right Bus) circuit breakers.

k) Check the antenna electrical bonding as per Para 3.D and/or Para 3.E.

l) Do a function test of the Transponder System, refer to Para 3.B.

5. Troubleshooting

A. Transponder System (KXP 2290 (Mod 6/2049) (Pre Mod 6/2354) or KXP 2290A (Post Mod 6/2354))

1) If the XPDR 1 and/or XPDR 2 is deemed inoperative check the following:

a) Ensure XPDR 1 (G3) and/or XPDR 2 (G4) circuit breaker is closed.

b) Check that the unit is securely installed and the retaining mechanism is securely tightened.

c) Visually inspect XPDR 1 and/or XPDR 2 connectors for damaged wires or contacts.

d) Visually inspect XPDR 1 and/or XPDR 2 antennas for corrosion and contaminants.

e) Inspect antenna external surfaces for damage.

f) Check that the antennas are properly mounted and screws are tight.

g) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

h) If repair or service is required, send to appropriate rated avionics repair facility.

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